

Page 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science concerned with building systems that can perform tasks that normally require human intelligence.

AI systems can process information, recognize patterns, understand language, make predictions, and support decision-making. Modern AI applications are used in search engines, recommendation systems, autonomous vehicles, healthcare, finance, and education.

Machine learning is one of the most important areas within AI. Machine learning algorithms learn patterns from data rather than relying entirely on manually written rules.

The main types of machine learning are supervised learning, unsupervised learning, and reinforcement learning. Each type is designed for different kinds of problems and data.

Page 2: Machine Learning

Supervised Learning

Supervised learning uses labeled examples to train a model. Classification predicts categories, while regression predicts numerical values. Examples include spam detection, house-price prediction, and credit-risk prediction.

Unsupervised Learning

Unsupervised learning works with data that does not have predefined labels. Clustering algorithms can group similar observations together. This can be useful for customer segmentation and exploratory data analysis.

Reinforcement Learning

In reinforcement learning, an agent interacts with an environment and receives rewards or penalties. Over time, it learns a policy that helps it choose actions that maximize long-term reward.

Page 3: Natural Language Processing and RAG

Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to work with human language. Common applications include translation, sentiment analysis, summarization, question answering, and chatbots.

Large Language Models

Large Language Models (LLMs) are neural network models trained on large amounts of text. They can generate text, answer questions, summarize documents, and assist with programming.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, connects a language model with external information. Documents can be loaded, split into chunks, converted into embeddings, and stored in a vector database. Relevant chunks are retrieved when a user asks a question.

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